2337.TW - Macronix (旺宏)

Full investment research | Model assumptions and source evidence

Owner valuation update - 15 October 2026

Tom authorizes a gradual re-rating: **6x in Q3 and Q4 2026, 7x in Q1 2027, 8x in Q2 2027 and 9x from Q3 2027 onward**. The multiple is selected by the ceiling quarter and applied to that quarter plus the next three operating quarters. The owner also extends the 6x starting multiple to Q3 2026. This supersedes the earlier Q3 exclusion; Q4 2026 and every later valuation quarter retain the previously approved schedule. This changes valuation only; all product revenue, costs, operating profit, probability/overlap treatment, attribution, shares, FX and Base scores remain unchanged.

Active Q4 2027 ceiling: TWD 226.36; unchanged by the Q3 extension. The active quarter applies 9x forward-four-quarter operating profit.

Ceiling quarter	Owner multiple	Current ceiling
Q3 2026	6x	TWD 156.08
Q4 2026	6x	TWD 158.83
Q1 2027	7x	TWD 182.86
Q2 2027	8x	TWD 206.53
Q3 2027	9x	TWD 229.46
Q4 2027	9x	TWD 226.36

This owner instruction supersedes the older constant-multiple convention wherever described in the retained research. Source dates and operating evidence remain at their original dates.

Framework 6.0 · Research cutoff: 6 September 2026

Status: Integrated original research and approved earnings model; evidence qualifications remain active.

Classification: TEMPORAL SPECIALTY-MEMORY BOTTLENECK - legacy supply allocation and qualification

Portfolio Excel Base: 75/100 - current saved master; the removed historical override is not reinstated. Analytical recommendation: **75/100 (45/60)**, advisory only.

Active Q4 2027 operating ceiling: TWD 226.36 per share. Q1 2028: TWD 223.27 per share. Financial amounts: TWD millions unless stated otherwise. Ceilings: TWD per share.

The ceiling is one conditional operating case. Probability is not multiplied into revenue, operating profit or the ceiling. Portfolio Base and the analytical category score are distinct; the latter does not overwrite the former.

1. Thesis, products and competitive position

SPECIALTY MEMORY SHARE GAINER. Macronix serves nonvolatile control/storage uses through NOR, specialty NAND/eMMC, ROM and foundry services. High-reliability qualification supports customer retention, but lower-density specialty memory is a different market from leading-edge enterprise NAND or HBM. No HBM, 300-layer enterprise SSD or exclusive AI-platform revenue is added.

Reviewed Q2 financials show revenue TWD 19,124.524M, GP 12,320.018M, opex 3,352.482M and OP 8,967.536M. The primary H1 note identifies 960.752M of Q2 inventory recovery. Excluding that recovery gives GP 11,359.266M and OP 8,006.784M; this bridge is displayed separately and does not rewrite the actual accounts. No inventory recovery is assumed in future product GP. Sources: MX-1, MX-3.

Long-lived products and supplier withdrawal

Macronix manufactures nonvolatile memory for applications where product longevity, reliability and qualification matter. NOR, SLC NAND and lower-density legacy memory can serve industrial, automotive and other embedded designs whose replacement cycles differ from leading-edge commodity memory. Existing customer qualification can make a supplier difficult to replace within a product's life.

The original thesis emphasizes larger suppliers reallocating resources toward more advanced products. This is a time-dependent allocation opportunity, not evidence that competitors can never return or that Macronix is the sole possible global source. Winbond, GigaDevice and other alternatives, customer redesign and larger-density substitution remain relevant.

Capacity, pricing and customer response

Supplier last-order or final-shipment dates in the original channel notes should be treated as dated observations until verified with the supplier or customer. End-of-life schedules can create inventory builds ahead of withdrawal and later normalization; they do not guarantee perpetual shortage pricing.

Macronix's captured demand depends on qualified capacity, equipment commissioning and customer order conversion. Gross profit should distinguish normal sales economics from inventory recoveries and the lag between selling prices and manufacturing costs. Higher-layer NAND and additional embedded/LEO applications are optionality until qualification and a distinct revenue bridge support them; their broad end-market TAM is not added wholesale to specialty-memory sales.

Research provenance: the product and competitive analysis in this section synthesizes the supplied original research. Dated market-share, process and channel claims retain their original evidence status; this integration is not a new independent verification of those claims. Current earnings and model assumptions are set out below.

2. Earnings, guidance and operating assumptions

Management guidance and capacity

The July presentation describes high-density NOR and SLC/eMMC shortages, adds planned capex 15.8B, and targets automotive eMMC shipments in 2027. Q2 operating cash flow 7.839B and capex 0.479B are actual cash figures; a capex budget is not completed equipment or revenue. NOR revenue growth greatly exceeded its flat sequential density index, reinforcing the need to separate price/mix from units. Source: MX-2.

The April and July primary presentations were reviewed in full. Public call pages provided summaries and only partial verbatim transcripts. Their capacity and layer-count details are treated as research leads, not certified guidance. Both complete call records remain an evidence gap. In particular, erroneous conflation of HBM

with QLC in the secondary summary is excluded. No universal fixed-price multi-year agreement is assumed; Micron contract protections cannot be transferred to this company.

Bottom-up revenue assumptions

Q3/Q4 2026 NOR sales are TWD 11/12B, NAND/eMMC11/13B, ROM2/1.5B and foundry 0.60/0.65B. These are successful shortage-cycle forecasts, not numerical company quarterly guidance. Calendar 2027-30 product sales in billions are NOR46/48/50/52, NAND50/60/69/76, ROM6/6.5/7/7.5 and foundry 2.6/2.8/3.0/3.2. Annual seasonality is 22/24/27/27%.

Against Q2 annualized product-mix approximations, 2027 NOR requires about 25% more revenue and NAND52% more. One explicit decomposition is NOR10% additional shipped bits ×14% ASP/mix and NAND25% bits ×22% ASP/mix. These factors are analytical and approximate because reported mix is rounded and parent based. For 2028, NAND revenue+20% could require 33% more bits with a 10% ASP reduction, while NOR+4% could require 10% more bits with a 5% ASP decline. Higher die density and equipment delivery must support those output requirements; no absolute wafer-capacity certification is claimed.

Later NAND sales growth slows 15/10% and NOR roughly 4% annually. The revenue case is capped by a specified product path, not a percentage of the entire AI infrastructure spend. NOR boot/control content, eMMC storage and ROM game content are separate demand sources.

Product economics and normalization

Q3/Q4 model GMs are NOR72%, NAND74%, ROM40% and foundry 23%; these assume further realized price/mix improvement without inventory recovery.2027-30 GMs are NOR67/60/52/48%, NAND68/58/50/46%, ROM40/40/38/36% and foundry 22%. This is an explicit cycle and expansion-cost normalization, not a probability haircut. Factory depreciation stays within GM.

Opex is 3,600/4,000M for the two 2026 model quarters. Thereafter quarterly fixed opex 2,600M grows 10% annually, plus 4% of sales for variable overhead, selling and incentives. This separates cost behavior from the legacy flat operating margin and provides room for R&D and compensation as revenue scales. Actual expense notes and the large expansion budget constrain the margin assumptions; exact project depreciation schedules remain unresolved.

Shares, scoring and next checkpoints

Use the verified Q2 diluted weighted-average 1,996.986M as a flat forward proxy through 2028, then 1% annual net dilution for 2029/30. This includes Q2 convertible/employee-compensation effects and is above the obsolete 1,930M assumption. It is not a promise of the future capital structure; financing and the next option/award reconciliation can change it. Source: MX-3.

Each ceiling is current-plus-next-three-quarter OP × the owner-approved quarter-specific multiple / period shares, with no probability factor. Watch monthly revenue, Q3 price-versus-bit growth, recovery-adjusted GP, equipment commissioning, depreciation and cash conversion. Shared learning: Longsys and Penguin purchase memory inputs, whereas Macronix manufactures specialty memory. Their gross-margin paths must respond differently to the same price cycle.

3. Analytical score

The category arithmetic gives **45/60 = 75/100** after the existing -4-point risk deduction. The current saved portfolio Excel Base is **75/100**, unchanged by this valuation instruction. Risk-class detail appears in section 7.

Category	Points	Max	Basis
Core thesis	16	20	High-reliability NOR and specialty NAND qualification support customer retention.
Cycle stage	9	10	Realized pricing, mix and earnings materially stronger; cyclical normalization still modeled.
TAM expansion	8	10	Specialty control memory and rising system content; not a claim to whole AI memory TAM.
Catalysts	5	5	Added capex, high-density NOR and automotive eMMC plan.
Macro	8	10	Supply reallocation supports specialty products; affordability and new supply constrain duration.
Alpha	3	5	Primary financial notes isolate inventory recovery from current product profit.
Risks	-4	0	Supply cycle, equipment timing, customer requalification, capex, execution and dilution.

4. Product economics and quarterly operating model

Forecasts distinguish reported anchors from guidance and analyst assumptions. Where the model uses segment OP or an adjusted profit proxy rather than product GP less opex, the stated basis applies; do not treat those proxies as a newly disclosed GAAP decomposition.

Vector evidence

Vector	Source / derivation	Confidence
NOR	Q2 reported parent product share 48%; density shipment index flat at 118 versus Q1 while revenue rose 48%	HIGH reported indicators; LOW future product GP
NAND including eMMC	Q2 parent share 43%; eMMC58% of NAND revenue; automotive shipment target 2027	HIGH direction/mix; LOW absolute forecast units
ROM	Q2 share 6%; cartridge/content demand differs from data-center memory	HIGH approximate mix; LOW future adoption
Foundry / other	Q2 share 3%; residual business kept distinct	HIGH mix; LOW forecast

Product revenue build

Quarter	NOR	NAND including eMMC	ROM	Foundry / other	Total
Q3 2026	11,000.00	11,000.00	2,000.00	600.00	24,600.00
Q4 2026	12,000.00	13,000.00	1,500.00	650.00	27,150.00
Q1 2027	10,120.00	11,000.00	1,320.00	572.00	23,012.00
Q2 2027	11,040.00	12,000.00	1,440.00	624.00	25,104.00
Q3 2027	12,420.00	13,500.00	1,620.00	702.00	28,242.00
Q4 2027	12,420.00	13,500.00	1,620.00	702.00	28,242.00
Q1 2028	10,560.00	13,200.00	1,430.00	616.00	25,806.00
Q2 2028	11,520.00	14,400.00	1,560.00	672.00	28,152.00
Q3 2028	12,960.00	16,200.00	1,755.00	756.00	31,671.00
Q4 2028	12,960.00	16,200.00	1,755.00	756.00	31,671.00
Q1 2029	11,000.00	15,180.00	1,540.00	660.00	28,380.00
Q2 2029	12,000.00	16,560.00	1,680.00	720.00	30,960.00
Q3 2029	13,500.00	18,630.00	1,890.00	810.00	34,830.00
Q4 2029	13,500.00	18,630.00	1,890.00	810.00	34,830.00
Q1 2030	11,440.00	16,720.00	1,650.00	704.00	30,514.00
Q2 2030	12,480.00	18,240.00	1,800.00	768.00	33,288.00
Q3 2030	14,040.00	20,520.00	2,025.00	864.00	37,449.00
Q4 2030	14,040.00	20,520.00	2,025.00	864.00	37,449.00

Product gross margins

Quarter	NOR	NAND including eMMC	ROM	Foundry / other
Q3 2026	72.0%	74.0%	40.0%	23.0%
Q4 2026	72.0%	74.0%	40.0%	23.0%
Q1 2027	67.0%	68.0%	40.0%	22.0%
Q2 2027	67.0%	68.0%	40.0%	22.0%
Q3 2027	67.0%	68.0%	40.0%	22.0%
Q4 2027	67.0%	68.0%	40.0%	22.0%
Q1 2028	60.0%	58.0%	40.0%	22.0%
Q2 2028	60.0%	58.0%	40.0%	22.0%
Q3 2028	60.0%	58.0%	40.0%	22.0%
Q4 2028	60.0%	58.0%	40.0%	22.0%
Q1 2029	52.0%	50.0%	38.0%	22.0%
Q2 2029	52.0%	50.0%	38.0%	22.0%
Q3 2029	52.0%	50.0%	38.0%	22.0%
Q4 2029	52.0%	50.0%	38.0%	22.0%
Q1 2030	48.0%	46.0%	36.0%	22.0%
Q2 2030	48.0%	46.0%	36.0%	22.0%
Q3 2030	48.0%	46.0%	36.0%	22.0%
Q4 2030	48.0%	46.0%	36.0%	22.0%

Quarterly operating model

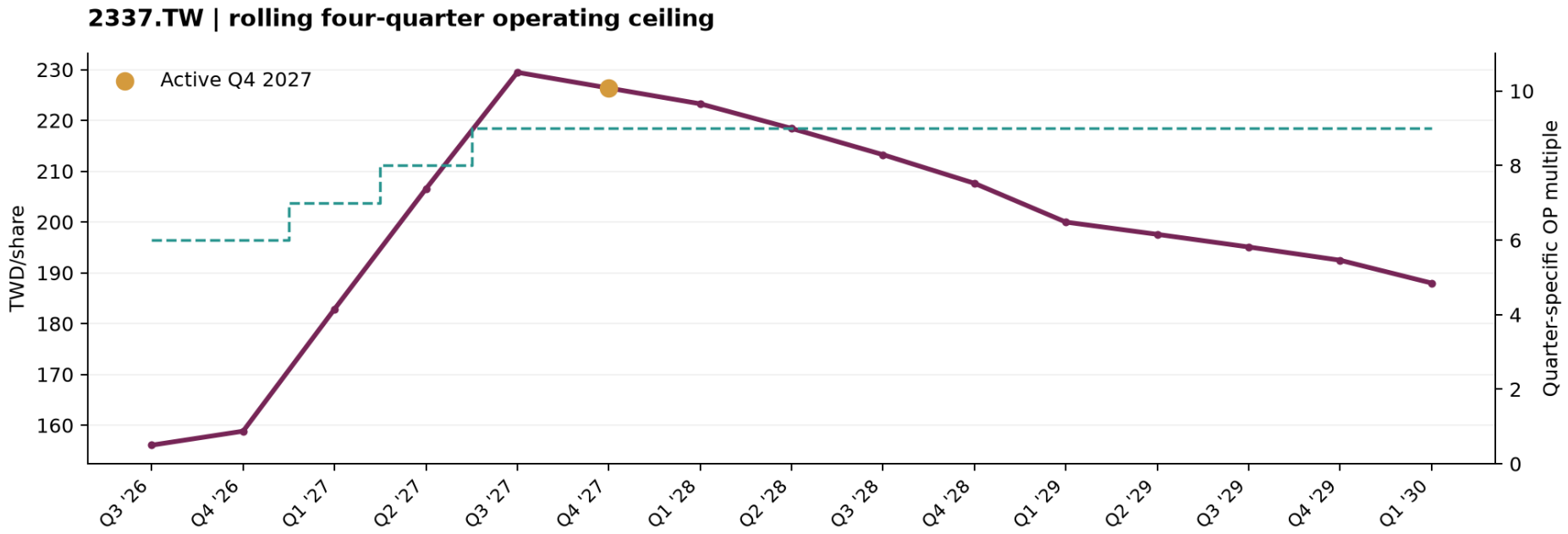
Quarter	Status	Revenue	Gross profit	Gross margin	Opex	Operating profit	OP margin	Forecast confidence
Q3 2026	Model	24,600.00	16,998.00	69.1%	3,600.00	13,398.00	54.5%	LOW; reported anchors distinguished above
Q4 2026	Model	27,150.00	19,009.50	70.0%	4,000.00	15,009.50	55.3%	LOW; reported anchors distinguished above
Q1 2027	Model	23,012.00	14,914.24	64.8%	3,780.48	11,133.76	48.4%	LOW; reported anchors distinguished above
Q2 2027	Model	25,104.00	16,270.08	64.8%	3,864.16	12,405.92	49.4%	LOW; reported anchors distinguished above
Q3 2027	Model	28,242.00	18,303.84	64.8%	3,989.68	14,314.16	50.7%	LOW; reported anchors distinguished above
Q4 2027	Model	28,242.00	18,303.84	64.8%	3,989.68	14,314.16	50.7%	LOW; reported anchors distinguished above
Q1 2028	Model	25,806.00	14,699.52	57.0%	4,178.24	10,521.28	40.8%	LOW; reported anchors distinguished above
Q2 2028	Model	28,152.00	16,035.84	57.0%	4,272.08	11,763.76	41.8%	LOW; reported anchors distinguished above
Q3 2028	Model	31,671.00	18,040.32	57.0%	4,412.84	13,627.48	43.0%	LOW; reported anchors distinguished above

Quarter	Status	Revenue	Gross profit	Gross margin	Opex	Operating profit	OP margin	Forecast confidence
Q4 2028	Model	31,671.00	18,040.32	57.0%	4,412.84	13,627.48	43.0%	LOW; reported anchors distinguished above
Q1 2029	Model	28,380.00	14,040.40	49.5%	4,595.80	9,444.60	33.3%	LOW; reported anchors distinguished above
Q2 2029	Model	30,960.00	15,316.80	49.5%	4,699.00	10,617.80	34.3%	LOW; reported anchors distinguished above
Q3 2029	Model	34,830.00	17,231.40	49.5%	4,853.80	12,377.60	35.5%	LOW; reported anchors distinguished above
Q4 2029	Model	34,830.00	17,231.40	49.5%	4,853.80	12,377.60	35.5%	LOW; reported anchors distinguished above
Q1 2030	Model	30,514.00	13,931.28	45.7%	5,027.22	8,904.06	29.2%	LOW; reported anchors distinguished above
Q2 2030	Model	33,288.00	15,197.76	45.7%	5,138.18	10,059.58	30.2%	LOW; reported anchors distinguished above
Q3 2030	Model	37,449.00	17,097.48	45.7%	5,304.62	11,792.86	31.5%	LOW; reported anchors distinguished above
Q4 2030	Model	37,449.00	17,097.48	45.7%	5,304.62	11,792.86	31.5%	LOW; reported anchors distinguished above

Annual model cross-check

Model year (quarter convention above)	Revenue	Operating profit	OP margin
2027	104,600.00	52,168.00	49.9%
2028	117,300.00	49,540.00	42.2%
2029	129,000.00	44,817.60	34.7%
2030	138,700.00	42,549.36	30.7%

5. Forward four-quarter operating ceiling



2337.TW current ceiling path and owner-approved gradual operating-profit re-rating, extended to Q3 2026. Gold marks active Q4 2027. Operating forecasts and share basis are unchanged.

Each row uses that quarter plus the next three quarters of operating profit × the sanctioned multiple ÷ that quarter's diluted shares. Probability is not multiplied into the ceiling. The final three operating quarters have no ceiling because a complete four-quarter window is unavailable.

Quarter	Exact earnings window	Forward four-quarter basis	Multiple	Share denominator (M)	Ceiling per share
Q3 2026	Q3 2026 → Q2 2027	51,947.18	6x	1,996.99	156.08
Q4 2026	Q4 2026 → Q3 2027	52,863.34	6x	1,996.99	158.83
Q1 2027	Q1 2027 → Q4 2027	52,168.00	7x	1,996.99	182.86
Q2 2027	Q2 2027 → Q1 2028	51,555.52	8x	1,996.99	206.53
Q3 2027	Q3 2027 → Q2 2028	50,913.36	9x	1,996.99	229.46
Q4 2027	Q4 2027 → Q3 2028	50,226.68	9x	1,996.99	226.36
Q1 2028	Q1 2028 → Q4 2028	49,540.00	9x	1,996.99	223.27
Q2 2028	Q2 2028 → Q1 2029	48,463.32	9x	1,996.99	218.41
Q3 2028	Q3 2028 → Q2 2029	47,317.36	9x	1,996.99	213.25
Q4 2028	Q4 2028 → Q3 2029	46,067.48	9x	1,996.99	207.62
Q1 2029	Q1 2029 → Q4 2029	44,817.60	9x	2,016.96	199.98
Q2 2029	Q2 2029 → Q1 2030	44,277.06	9x	2,016.96	197.57
Q3 2029	Q3 2029 → Q2 2030	43,718.84	9x	2,016.96	195.08
Q4 2029	Q4 2029 → Q3 2030	43,134.10	9x	2,016.96	192.47
Q1 2030	Q1 2030 → Q4 2030	42,549.36	9x	2,037.13	187.98

Mechanical share sensitivity

Holding the Q1 2028 operating case, attribution and currency assumptions fixed:

Economic denominator	TWD ceiling per share
Approved denominator	223.27
5% more shares	212.63
10% more shares	202.97

This is a denominator sensitivity, not a second operating scenario or a forecast of issuance.

6. Industry relationships and read-throughs

Macronix is a manufacturer; ESMT and GigaDevice depend on external manufacturing for relevant products, while Longsys and Penguin purchase memory inputs. The same shortage can raise selling prices for one business and compress replacement-cost margins for another.

7. Risks and falsification tests

The following ledger makes the existing aggregate risk deduction explicit. It does not introduce another deduction or change the analytical score.

Risk class	Existing deduction
Supply-cycle reversal and selling prices	-1
Equipment commissioning, depreciation and capex	-1
Customer requalification and product substitution	-1
Execution, cash conversion and dilution	-1
Total	**-4**

Numbered monitoring tests

#	Checkpoint	Evidence supporting the operating case	Counter-test / failure signal
1	Price versus bits	Separate monthly/quarterly growth into shipments, mix and pricing.	An inventory build or price spike does not prove sustainable bit demand.
2	Supply withdrawals	Verify specific product and shipment schedules.	Delayed exits or competitive re-entry weakens the scarcity assumption.
3	Capacity	Reconcile commissioned equipment and usable output to the forecast.	Capex announcements without yield and customer approval do not validate sales.
4	Normalized profit	Track recovery-adjusted GP, depreciation and cash generation.	Inventory gains without replacement-cost profitability do not support the recurring margin.

8. Sources, evidence gaps and model snapshot

Sources below are those documented in the supplied earnings review. LOW-confidence product allocations, future margins, capacity bridges and incomplete management-call records remain unresolved where stated. Integrating the original research does not itself establish full Framework 6.0 evidentiary certification.

- **MX-1** · 2026-07-28 · 2026第二季營運報告. Primary actuals and product mix.
- **MX-2** · 2026-07-28 · 2026第二季法說會簡報. Capex, high-density NOR, NAND/eMMC and product shipment indices.
- **MX-3** · 2026 H1; available by cutoff · 115年第2季合併財務報告. Exact accounts, inventory recovery and diluted EPS denominator.
- **MX-4** · 2026-04-27 · 2026第一季法說會簡報. Previous management presentation and starting product trajectory.
- **MX-5** · 2026-07-28 · Q2 call summary / partial transcript. Secondary AI transcription aid only; unsupported technical statements excluded.
- **MX-6** · 2026-04-27 · Q1 call summary / partial transcript. Prior-period qualitative comparison; not a fully reviewed verbatim call.
- **MX-7** · 2026-09-01 · Original model and framework - User upload:2337_TW_Q_Model. Historical 15x OP convention, superseded from Q3 2026 by the October 15 owner schedule and explicit Q3 extension. Original separate artifact not included.

The unchanged original source document is retained separately for full research provenance. It is an archive, not an additional active valuation case. Superseded score/ceiling tables are not incorporated into this active artifact.

Model snapshot: approved portfolio operating case at 6 September 2026; 15 ceiling endpoints from Q3 2026 through Q1 2030. Active Q4 2027 = TWD 226.361186 per share; Q1 2028 = TWD 223.266463 per share. Owner valuation schedule updated October 15, 2026; operating research cutoff retained. Portfolio Excel Base = 75/100, matching the current master; no Base change in this run. Category score = 45/60. Forecast evidence and accounting exceptions are stated above.